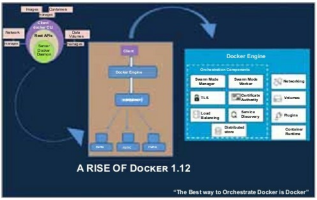

Figure 1: The evolution of Docker 1.12

discovery, Raft-based consensus, task scheduling and much more. Let's look at the features Docker Swarm Mode adds to Docker Cluster functionality, as shown in Figure 2.

Looking at these features, Docker Swarm Mode brings the following benefits.

- **Distributed:** Swarm Mode uses the Raft Consensus algorithm in order to coordinate, and does not rely on a single point of failure to make decisions.
- **Secure**: Node communication and membership within a Swarm are secure out-of-the-box. Swarm Mode uses mutual TLS for node authentication, role authorisation, transport encryption, and for automating both certificate issuance and rotation.
- **Simple**: Swarm Mode is operationally simple and minimises infrastructure dependencies. It does not need an external database to operate. It uses the internally distributed State store.

Figure 3 depicts Swarm Mode cluster architecture. Fundamentally, it's a master and slave architecture. Every node in a swarm is a Docker host running a Docker engine. Some of the nodes have a privileged role called the manager. The manager node participates in the 'Raft consensus' group. As shown in Figure 3, components in blue share an internal distributed state store of the cluster, while the green coloured components/boxes are worker nodes. The worker nodes receive work instructions from the manager

Figure 2: Features of Docker Swarm Mode

group, and this is clearly shown in dashed lines.

Getting started with Docker Engine 1.12

In this section, we will cover the following aspects:

- Initialising the Swarm Mode
- Creating the services and tasks
- Scaling the service

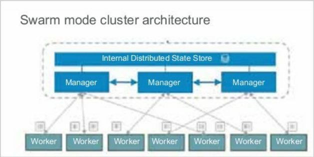

Figure 3: Swarm Mode Cluster architecture

- Rolling updates
- Promoting a node to the manager group

To test drive the Docker Mode, I used a four-node cluster in the Google Cloud Engine — all running the latest stable Ubuntu 16.04 system as shown in Figure 4.

Initialising the Swarm Mode

Docker 1.12 is still in the experimental phase. Setting up Docker 1.12-rc2 on all the nodes should be simple enough with the following command:

```
#curl -fsSL https://test.docker.com/ | sh
```

Name	Zone	Network	In use by	In use IP	External IP	Connect
swarm-agent3	asia-east1-b	default		10.140.0.2	130.211.249.130	SSH
swarm-agent1	asia-east1-c	default		10.140.0.3	130.211.251.130	SSH
swarm-agent2	asia-east1-c	default		10.140.0.4	130.199.169.202	SSH
swarm-agent1	asia-east1-b	default		10.140.0.5	104.199.164.86	SSH

Figure 4: Test set-up

Run the command (as shown in Figure 5) to initialise Swarm Mode under the master node.

Figure 5: Initialising Swarm Mode

Listing of the Docker Swarm master node is shown in Figure 6.

Figure 6: Listing the master node